

Hack The World(s) – Track 7

Learned Cost / Value for Planning

Team HTW

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1 Goal

The Track 7 hypothesis is that distance in latent space is a crude planning cost. A JEPA world model can predict future latent states, but the standard MPC objective,

$$J_{\text{dist}}(a_{1:H}) = \sum_{t=1}^H \|\hat{z}_t - z_g\|_2^2,$$

does not necessarily correlate with task success. In Two Rooms, the optimal trajectory is often non-monotonic: the agent must first move toward the door, which can temporarily increase Euclidean or latent distance to the final goal.

We replace this hand-crafted distance with a learned goal-conditioned value function:

$$V_\psi(z_t, z_g) \in (0, 1),$$

trained on the world model’s own rollouts. At planning time MPPI minimizes

$$J_{\text{value}}(a_{1:H}) = \sum_{t=1}^H \gamma^{t-1} (1 - V_\psi(\hat{z}_t, z_g)).$$

The planner therefore optimizes a quantity trained to predict reachability rather than raw representation similarity.

2 Why This Fits the Track

The official track asks for three concrete changes:

- a new MPC objective in `eb_jepa/planning.py`, registered in `objective_name_map`;
- a small value head trained alongside, or on top of, the JEPA;
- metrics on Two Rooms success and planning compute versus the distance baseline, especially on hard non-monotonic episodes.

The implementation already contains the core hook: `LearnedValueMPCObjective` is registered as `learned_value`, and `GoalValueHead` maps (z_t, z_g) to a scalar value. The next run should make this the main experiment rather than an auxiliary maze result.

3 Lessons from the Maze README

The Maze README is important context, but it should not replace the Track 7 Two Rooms experiment. It teaches four design constraints:

- **Freeze a proven world model first.** Maze co-training lowered subgoal MSE but hurt success, because moving the encoder degraded the wall-aware dynamics. Our main Track 7 comparison therefore trains the value head on a frozen WM before attempting any joint fine-tuning.
- **Keep the comparison controlled.** The Maze value result is strong only when the planner scaffolding is matched: same WM, same planner, same episodes, only the cost changes. We copy exactly that protocol for Two Rooms.
- **Do not hide oracle guidance in the planner.** Maze A* waypoints are useful as a stress-test, but the Track 7 story should not depend on A*. A* can be an oracle for analysis, not the source of the main action signal.
- **Horizon matters.** Maze greedy-global planning stays at 0% because a value trained on short windows cannot guide long-range navigation. Two Rooms is the right first target because its non-monotonic door crossing is hard enough to expose the flaw of latent distance while remaining within the value’s rollout horizon.

Thus the main Track 7 run is not “Maze hierarchy again”. It is the cleaner learned-cost experiment requested by the track, with Maze kept as supporting evidence and an ambitious extension.

4 Training Objective

We train a value head with a TD-MPC-style bootstrap. For each sampled trajectory window, the hindsight goal is the final encoded frame $z_g = z_T$. The target is

$$y_t = r_t + \gamma(1 - d_t)V_{\bar{\psi}}(z_{t+1}, z_g),$$

where $V_{\bar{\psi}}$ is an EMA target network, $r_t = 1$ only at the endpoint, and d_t marks the terminal endpoint transition.

The loss is computed on both encoded real latents and imagined world-model rollout latents:

$$\mathcal{L}_{\text{value}} = \|V_{\psi}(z_t, z_g) - y_t\|_2^2 + \|V_{\psi}(\hat{z}_t, z_g) - y_t\|_2^2.$$

The second term matters: MPPI will query the value on imagined states, so the head must be calibrated on the distribution produced by the world model.

5 Controlled Comparison

The main experiment changes exactly one thing: the planning objective.

same world model, same MPPI, same seeds, same episodes

`repr_dist` vs. `learned_value`.

Condition	Objective	Purpose
Baseline	<code>repr_dist</code>	Official distance-in-latent-space baseline
Ours	<code>learned_value</code>	TD-MPC value learned from WM rollouts
Oracle report only	A* / shortest path	Upper bound, not used for training in main result

Table 1: Track 7 comparison. A* is only an analysis oracle in the main story.

6 Metrics

We report:

- success rate on Two Rooms;
- mean final distance to goal;
- average episode wall time and approximate world-model calls;
- success on the hard subset: start and goal on opposite sides of the wall, normal-level layout, long door-seeking trajectories;
- qualitative GIFs: one success, one failure, one non-monotonic case.

For a compute story, we run at least two MPPI budgets:

`num_samples` $\in$ $\{50, 200\}$, `n_iters` = 20, `plan_length` = 90.

This gives a small Pareto curve: success versus planning time.

7 Expected Jury Story

The concise claim should be:

The baseline world-model planner optimizes latent similarity, but latent similarity is not a task cost. We train a TD-MPC-style value head on the JEPA world model’s own rollouts and use it as the MPPI objective. On non-monotonic Two Rooms episodes, this value provides a reachability signal that is better aligned with success than raw latent distance.

8 Execution Checklist

1. Train or reuse a stable Two Rooms world model checkpoint.
2. Train `GoalValueHead` on the frozen world model using `train_value.yaml`.
3. Evaluate `repr_dist` and `learned_value` on the same `eval_track7.yaml` episodes.
4. Repeat at low and standard MPPI compute budgets.
5. Export table, Pareto plot, and 2–3 GIFs.
6. If time remains, run an ambitious joint fine-tune with `freeze_world_model=false` and small `value_coeff`.

9 Risk Management

If the learned value underperforms globally, we still have a useful result: the value can be analyzed by horizon. A short-window TD target may improve near-goal decisions while failing on long-range navigation. This is a clean negative or partial result and directly supports the paper’s stated limitation: latent-distance costs are not enough, but learned values need horizon-aware training.